

BEATCUBES

MUSICAL INSTRUMENT



IMPACT OF MUSIC ON AN INDIVIDUALS

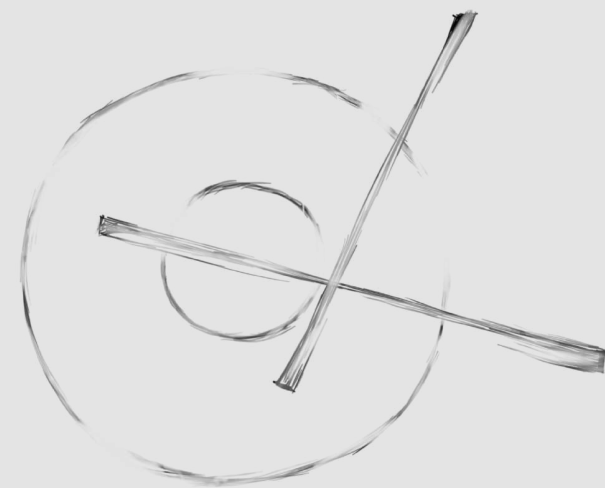
Listening To Music Creates Feeling A Sense Of Relaxation, Joy, Or Excitement.

Music Can Evoke Memories Or Emotions, Providing A Sense Of Comfort Or Nostalgia For The Listener.

In Some Cases, It Can Even Have Therapeutic Effects, Helping To Reduce Stress, Anxiety, Or Pain.

Experience Of Listening To Music Can Be A Deeply Personal And Emotional One, With The Potential To Positively Impact One's Mood And Well-Being.

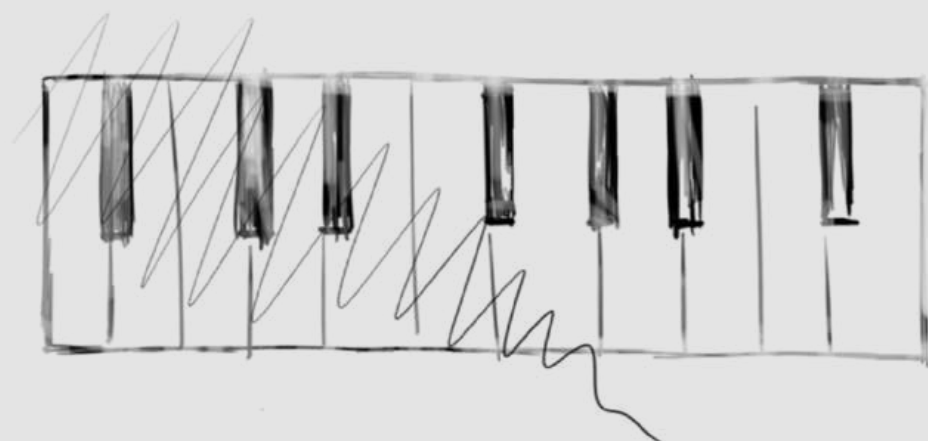




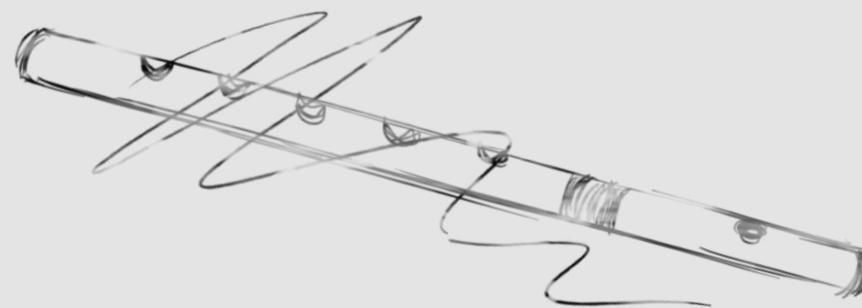
Drums produce sound through stretched material over a hollow body,



String instruments work by initiating vibrations in the strings,

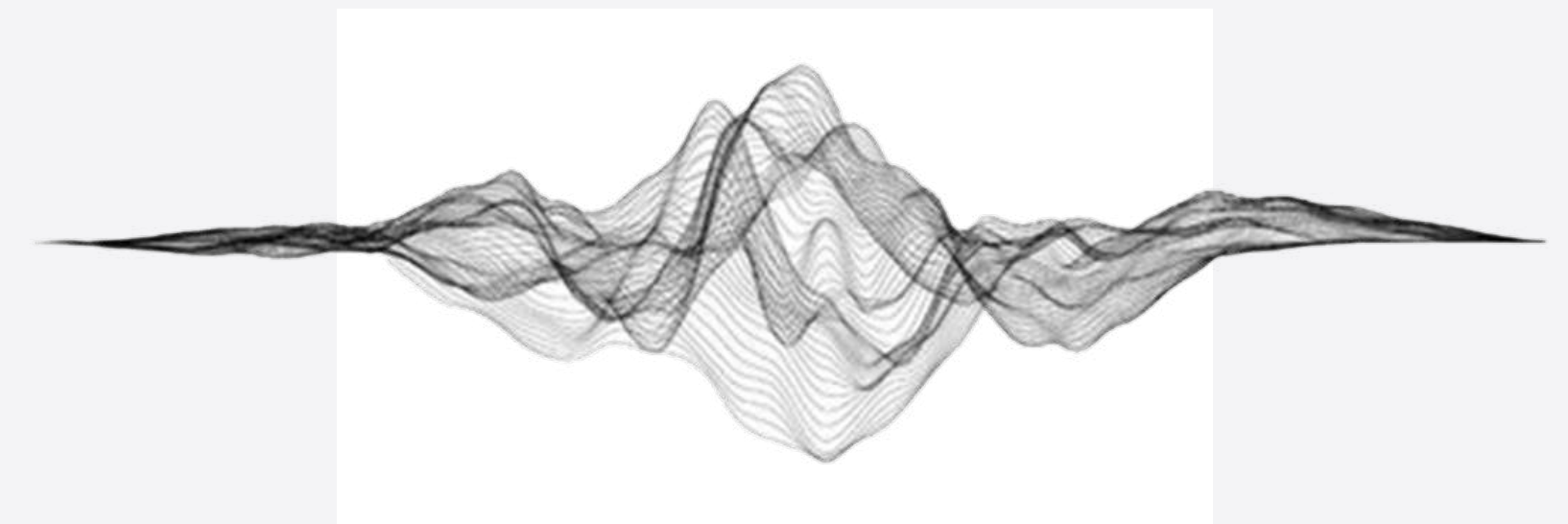


Creating sound by vibrating the air as the entire piano resonates



sound originates from a vibrating column of air

ROLE OF MUSICAL INSTRUMENTS

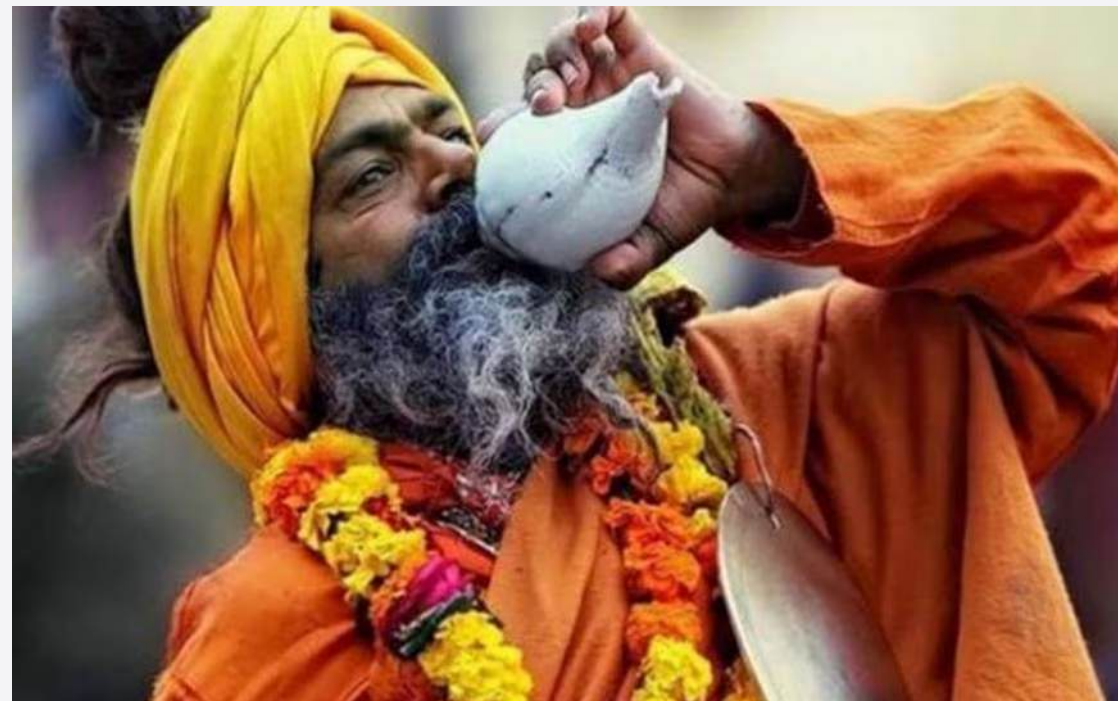


Musical instruments create sounds through striking, plucking, bowing, blowing, or vibrating. Used in various genres, solo or in groups, they come in diverse shapes and sizes. Examples include guitar, violin, and drums. Essential for music creation, instruments have been crucial in human culture for centuries.

EXPLORATION

During the research we explored several types of musical instrument (India) so, for our better understanding we classified them into two main categories considering Professional/technically Hard to play and unprofessional/technically Easy to play

VADYA



We classified these instruments in naturally found, Tribal, spiritual, and traditional musical instruments

INSIGHTS

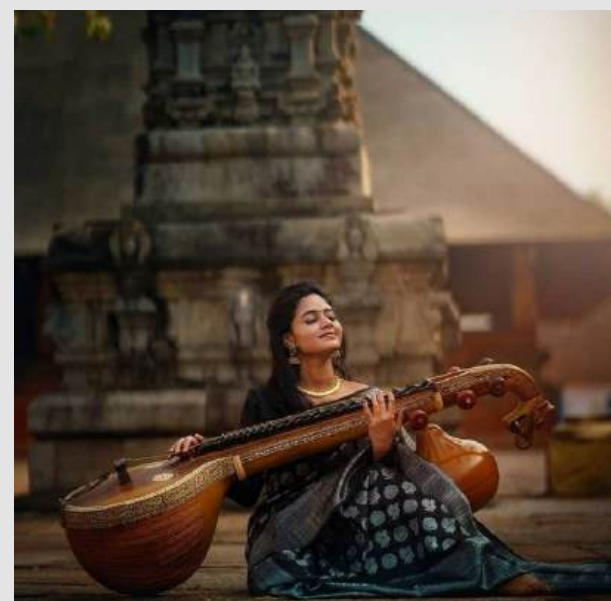
Natural musical instruments evoke peace, some believed to have healing effects on mind.

Spiritual instruments bring calmness, spreads energy, and creates positive environment.

Tribal instruments are dedicated to a particular tribe and are made by using the available resources around them.

Every tradition have their own traditional instruments, reflecting history and fostering pride.

KALAVADYA



This category includes Indian classical musical instruments and western musical instruments

INSIGHTS

These instruments are finely made to produce many different sounds.

Mastering these instruments demands music knowledge and Technical training.

This individual instrument have the potential to influence once different types of mood.

Every tradition have their own traditional instruments, reflecting history and fostering pride.

USER STUDY



Do you noticed?

We all have a habbit of tapping our hands or feet, often unconsciously, either out of boredom or to sync with the music vibe.

Observations in Human Behaviour

Tapping or drumming your fingers on an available surface (or fidgeting in another similar way) clearly demonstrates that you are growing bored or impatient.

Tapping has been clinically proven to manage stress, reduce cortisol levels, improve anxiety and even treat PTSD in veterans. “Because tapping is a gentle and easy-to-follow technique, anyone struggling with stress, anxiety or pain can benefit from it,” says Ortner.12-Aug-2022

Conclusion

To start playing the instrument, no prior knowledge or skill is required. Everyone with the desire to play may access it without spending years gaining experience on a musical instrument.



DESIGN BRIEF

To design a musical instrument which utilizes an unconscious human habit and contributes to the musical experience of any user.

DESIGN CONSIDERATIONS

Acoustics: Understanding the physics of sound and how it interacts with the materials used in instrument construction.

Materials: Selecting and working with materials such as wood, metal, plastics, and composites to create instruments with desired characteristics.

Ergonomics: Designing instruments that are comfortable and efficient to play, taking into account the physical abilities and limitations of the performer.

Aesthetics: Creating instruments that are visually appealing and reflect the personality and style of the musician.

Cultural and historical context: Considering the cultural and historical significance of different instrument designs and incorporating traditional elements into modern instruments.

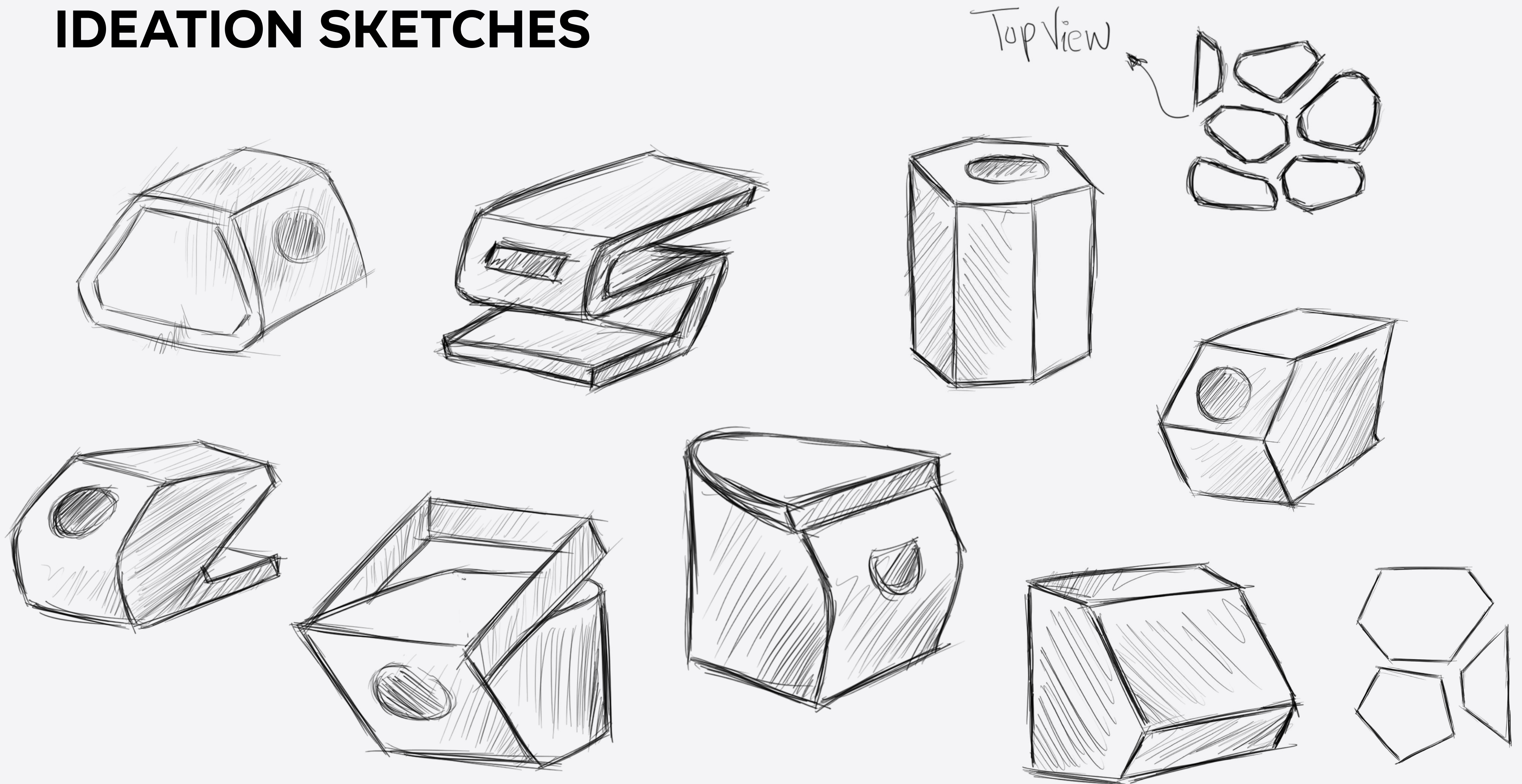
Playability and Sound Quality: Ensuring that the instruments are easy to play and produce high-quality sounds that meet the expectations of musicians and audiences.

FORM EXPLORATION

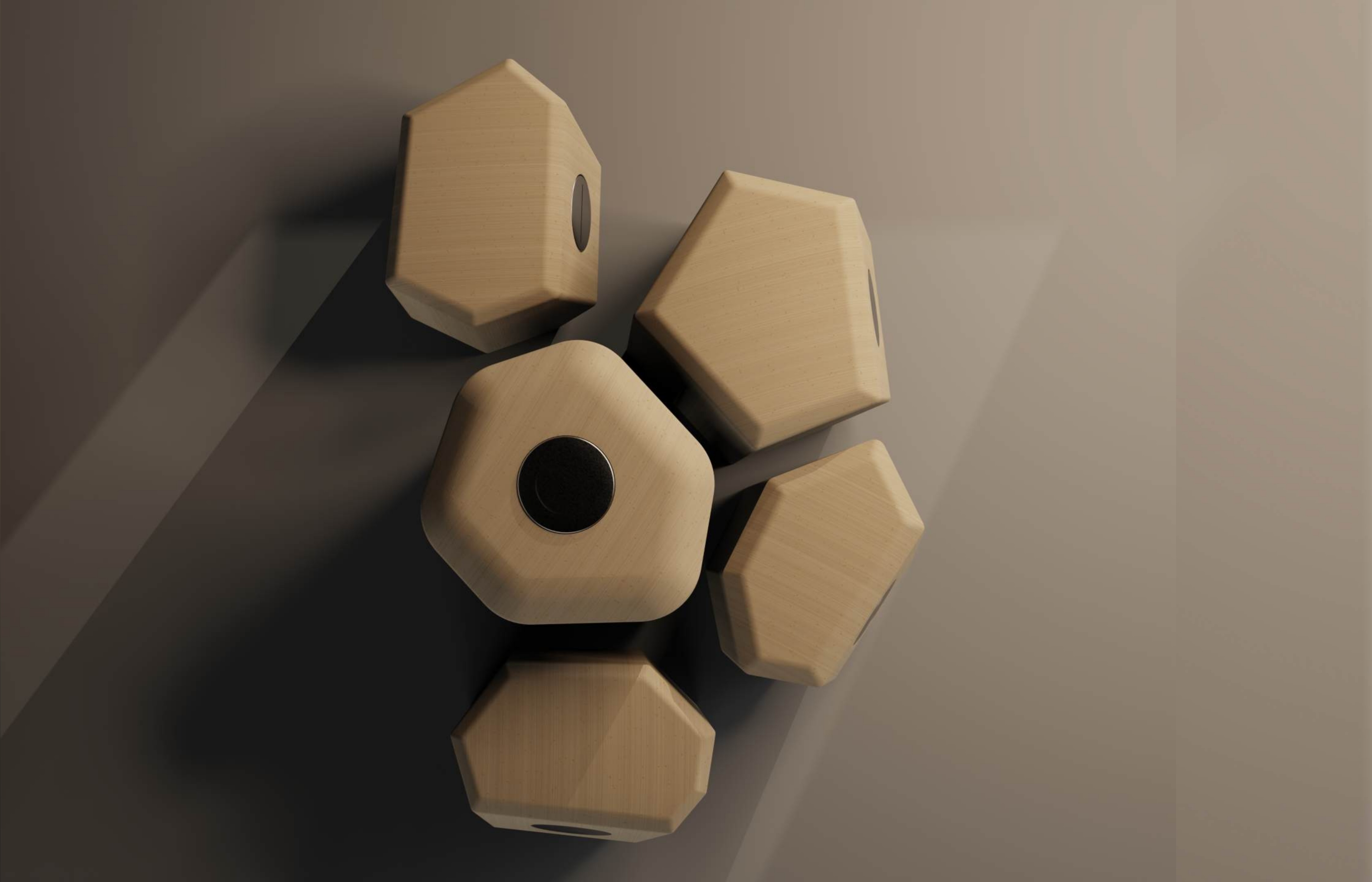


Contoured, Expressive, Easily Implementable, Modular, Perceptible, Exuberant

IDEATION SKETCHES



3D MODEL



HUMAN INTERACTION







FINAL PROTOTYPE



USER TESTING

